B.Tech (5th Semester) Computer Engg, Honors (2025)

DATA ANALYTICS SESSIONAL – 1 (CSH – 511)

Time : 1 Hour **Max. Marks : 15 (5+3+3+4)**

Q.1. a) If a supermarket has 1000 items then how many potential association rules can be generated?
b) Why support in Sequence Mining is lower than Association rule mining?
c) If we only keep closed itemsets, can we generate all frequent itemsets from it with its support? What if we store only max itemsets?
d) Suppose Node A has highest degree centrality and Node B has highest closeness centrality. What does this mean in terms of the role they play in the network?

Q.2. Consider the following sequences
T1: {A, B, D}, T2: {B, C, E}, T3: {A, B, C, E}, T4: {B, E}, T5: {A, B, C, E}, T6: {A, B, C, D, E}
T7: {C, D, E}. Construct FP- Tree and find all frequent patterns with support = 2.

Q.3. Given the DNA sequence database find all frequent sequences using minsup=4. Show the steps.

S1 : AATACAAGAAC; S2 : GTATGGTGAT; S3 : AACATGGCCAA; S4 : AAGCGTGGTCAA

Q.4. Given the following graph. Calculate the betweenness and closeness centrality.

